

Notes

December 21, 2023

Abstract

1 Selective inference

In Lemma 4 in the draft, we have derived the law under H_0 of

$$Z_{r,A} = \frac{n_A + 1}{n_A - 1} \sum_{i: X_i \in A} (Y_i - \bar{Y}_A)^2 + \frac{n_B + 1}{n_B - 1} \sum_{i: X_i \in B} (Y_i - \bar{Y}_B)^2 - r \frac{n + 1}{n - 1} \sum_{i=1}^n (Y_i - \bar{Y})^2$$

giving rise to $\mathcal{L}_0(R_A)$ for any fixed $A \subset D$. Let

$$\varphi((X_i, Y_i)_{i=1}^n) = \operatorname{argmin}_A \left(\sum_{i: X_i \in A} (Y_i - \bar{Y}_A)^2 + \sum_{i: X_i \in B} (Y_i - \bar{Y}_B)^2 \right)$$

an **MSE**-minimizing splitting rule (noting that $B = D \setminus A$ is determined by A). We want to find

$$\mathcal{L}_0(Z_{r,A} \mid \varphi = A) \neq \mathcal{L}_0(Z_{r,A}), \quad (1)$$

since $\{\varphi((X_i, Y_i)_{i=1}^n) = A\}$ provides additional information on the trainings sample $(X_i, Y_i)_{i=1}^n$. First note that if iid test data $(X_i^*, Y_i^*)_{i=1}^n$ is used to compute $Z_{r,A}$, then

$$\frac{\mathbb{P}_0(Z_{r,A}((X_i^*, Y_i^*)_{i=1}^n) \leq z, \varphi((X_i, Y_i)_{i=1}^n) = A)}{\mathbb{P}_0(\varphi((X_i, Y_i)_{i=1}^n) = A)} = \mathbb{P}_0(Z_{r,A}((X_i^*, Y_i^*)_{i=1}^n) \leq z), \quad (2)$$

which gives equality in (1). If, on the other hand, the same trainingsdata is used for φ and $Z_{r,A}$,

$$\mathcal{L}_0(Z_{r,A} \mid \varphi = A) \quad (3)$$

$$= \mathcal{L}_0 \left(\frac{n_A + 1}{n_A - 1} \sum_{i: X_i \in A} (Y_i - \bar{Y}_A)^2 + \frac{n_B + 1}{n_B - 1} \sum_{i: X_i \in B} (Y_i - \bar{Y}_B)^2 - r \frac{n + 1}{n - 1} \sum_{i=1}^n (Y_i - \bar{Y})^2 \right) \quad (4)$$

$$\left| \operatorname{argmin}_A \left(\sum_{i: X_i \in A} (Y_i - \bar{Y}_A)^2 + \sum_{i: X_i \in B} (Y_i - \bar{Y}_B)^2 \right) = A \right| \quad (5)$$

$$\approx \mathcal{L}_0 \left(\min_A \left(\frac{n_A + 1}{n_A - 1} \sum_{i=1}^{n_A} (Y_i - \bar{Y}_A)^2 + \frac{n_B + 1}{n_B - 1} \sum_{i=n_A+1}^n (Y_i - \bar{Y}_B)^2 \right) - r \frac{n + 1}{n - 1} \sum_{i=1}^n (Y_i - \bar{Y})^2 \right), \quad (6)$$

where the minimum runs over $n - 2m + 1$ possible splitpoints (here, m is the minimum number of sample points per region) of the ordered covariates $X_{(1)}, \dots, X_{(n)}$. The approximation in the last line is justified because for m sufficiently large, the factors in front of the sums are close to one. Moreover, the term

$$r \frac{n + 1}{n - 1} \sum_{i=1}^n (Y_i - \bar{Y})^2$$

is independent of the conditioning event. For the simulation in Figure 1, first trainingsdata $(X_i, Y_i)_{i=1}^n$ is generated with constant mean function (H_0 holds) for the computation of an **MSE**-minimizing splitpoint $\varphi((X_i, Y_i)_{i=1}^n) = A$. The purple bars correspond to the conditional distribution (3) (where $Z_{r,A} = Z_{r,A}((X_i, Y_i)_{i=1}^n)$). The orange bars are drawn from (6). One can see that the distributions are indistinguishable (here, the minimum number of sample points per region is $m = 10$). Just to compare, the green bars are generated using the optimal splitpoint $\varphi = A$, but computing $Z_{r,A} = Z_{r,A}((X_i^*, Y_i^*)_{i=1}^n)$ with independent data as in (2) giving rise to the null distribution from Lemma 4.

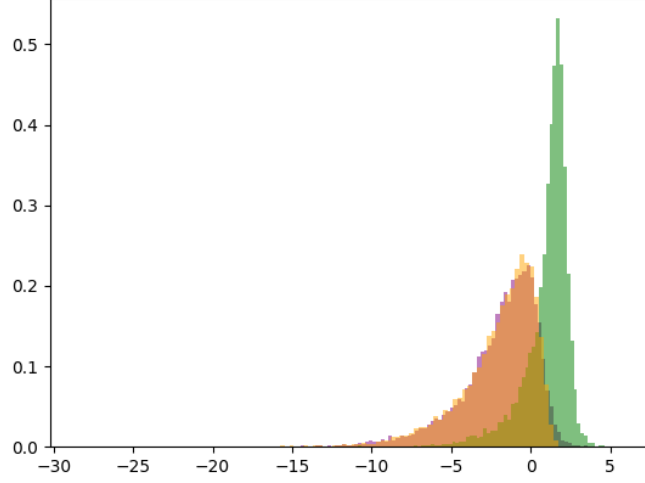


Figure 1:

The approximation in (6) above means that it is enough to determine the null distribution function $F_{\min R}^0(r) = \mathbb{P}_0(\min_A Z_{r,A} \leq 0)$ from (6) in order to obtain correct selective inference-adjusted p-values.

Let $(U_1, \dots, U_n)$ be iid standard normal and fix. From the proof of Lemma 4 it follows that under H_0 ,

$$\min_A Z_{r,A} \stackrel{d}{=} \min(V_k)_{k=m}^{n-m},$$

where

$$\begin{aligned} V_k \stackrel{d}{=} & \lambda_k^r \left(\sum_{i=1}^k U_i^2 - \frac{1}{k} \left(\sum_{i=1}^k U_i \right)^2 \right) + \lambda_{n-k}^r \left(\sum_{i=k+1}^n U_i^2 - \frac{1}{n-k} \left(\sum_{i=k+1}^n U_i \right)^2 \right) \\ & + \frac{\lambda_{\text{neg}}^r}{k(n-k)n} \left((n-k) \sum_{i=1}^k U_i - k \sum_{i=k+1}^n U_i \right)^2 \end{aligned}$$

with

$$\lambda_k^r = \frac{k+1}{k-1} - r \frac{n+1}{n-1}, \quad \lambda_{\text{neg}}^r = -r \frac{n+1}{n-1}.$$

Since all V_k are functions of the same random vector $(U_1, \dots, U_n)$, this de-

termines the joint distribution of $(V_k)_{k=m}^{n-m}$. Hence,

$$\begin{aligned} F_{\min R}^0(r) &= 1 - \mathbb{P}_0(V_m > 0, \dots, V_{n-m} > 0) \\ &= 1 - \int_{\mathbb{R}^n} \prod_{k=m}^{n-m} 1_{\{V_k(u_k) > 0\}} \varphi(u_1) \cdots \varphi(u_n) du_1 \dots du_n, \end{aligned}$$

where φ is the standard normal density.

It seems hard to compute this integral numerically, in particular for large n . However, approximations seem to work reasonably, cf. Figure 2.

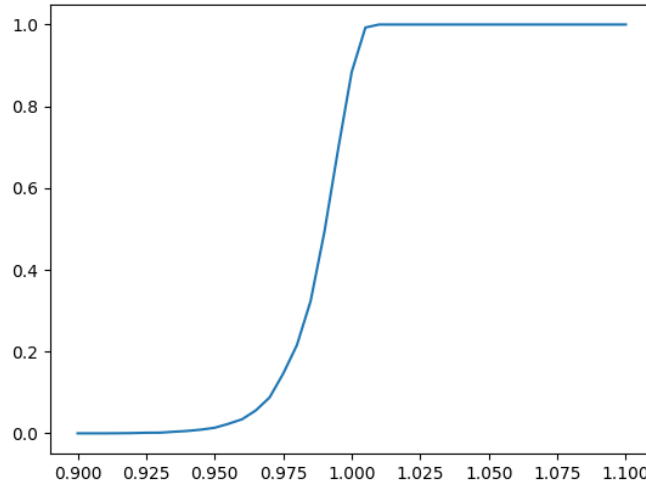


Figure 2: $F_{\min R}^0(r)$ for $n = 200$.

One approach could be to compute graphs of $F_{\min R}^0(r)$ like in Figure 2 for $n = 2m, \dots, 400$ and save the results in an extra file. Then, the algorithm could pick the right value from the external file (similar to a table of standard normal quantiles at school). For $n \geq 400$, it seems ok to use $1_{r>1}(r)$ as an approximation of $F_{\min R}^0(r)$.

One advantage of this approach is that it works the same if the residuals follow any other distribution. In this case, one just needs to replace $U_1, \dots, U_n$ from above by a centered and standardized iid vector of the new residual distribution (however, squared loss is necessary).